

Ex. No.: 10b)

Date: 10/04/2025

FIRST FIT

Aim:

To write a C program for implementation memory allocation methods for fixed partition using first fit.

Algorithm:

1. Define the max as 25.
2. Declare the variable frag[max], b[max], f[max], i, j, nb, nf, temp, highest=0, bf[max], ff[max].
3. Get the number of blocks, files, size of the blocks using for loop.
4. In for loop check bf[j] != 1, if so temp = b[j] - f[i]
5. Check highest

Program Code:

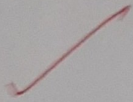
```
#include <stdio.h>

int main() {
    int b[] = {100, 45, 33, 45, 70};
    int pro[] = {20, 30, 50, 40, 10};
    int frag[5], flag[5];
    for (int i=0; i<5; i++) {
        frag[i] = 0;
        flag[i] = 0;
    }
    for (int i=0; i<5; i++) {
        for (int j=0; j<5; j++) {
            if (pro[i] < b[j] && flag[j] == 0) {
                frag[j] = b[j] - pro[i];
                flag[j] = 1;
                break;
            }
        }
    }
}
```

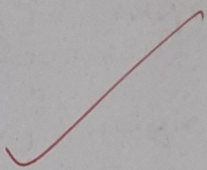


```
printf ("the fragments of the blocks are:\n");  
for (int i=0; i<5; i++)  
    printf ("%d\n", frag[i]);
```

y



Process NO	Process-size	Block-no	Fragment
P1	20	1	80
P2	20	2	15
P3	50	5	20
P4	40	4	5
P5	10	3	93



Sample Output:

```
Enter the number of blocks:4
Enter the number of files:3

Enter the size of the blocks:-
Block 1:5
Block 2:8
Block 3:4
Block 4:10
Enter the size of the files:-
File 1:1
File 2:4
File 3:7
```

File_no:	File_size :	Block_no:	Block_size:	Fragment
1	1	1	5	4
2	4	2	8	4
3	7	4	10	3_

OUTPUT:

The fragments of the blocks are :

80
15
23
5
20

Result:

Thus the C program for implementation memory allocation methods for fixed partition using first fit has been executed successfully